IT and Digital Culture Tutorial

Part 1: Exercises

Exercise 1: Numbers and strings (Chapter 3)

Without running the code, predict the result of each line, then check:

```
print(17 / 3, 17 // 3, 17 % 3, 2 ** 7)
word = "Python"
print(word[0], word[-1], word[2:5], word[:2] + word[2:], word[::-1])
```

Exercise 2: Lists (Chapters 3 and 5)

Create a list of the squares of 1 to 5, add 36 to the end, then show the last three items.

Exercise 3: Fibonacci function (Section 4.6)

Write `fib(n)` that returns a list of the Fibonacci numbers below `n`. Test it with `fib(50)`.

Exercise 4: Primes with for ... else (Section 4.4)

Print the prime numbers from 2 to 14 using a `for` loop with an `else` clause.

Exercise 5: The mutable default argument trap (Section 4.7)

This function should return a fresh list on each call. Explain what goes wrong and fix it.

```
def add_item(item, box=[]):
    box.append(item)
    return box

print(add_item(1), add_item(2))
```

Exercise 6: Dictionaries (Section 5.5)

Start with `tel = {"jack": 4098, "sape": 4139}`. Add `guido: 4127`, delete `sape`, list the keys in sorted order, and test whether `jack` is present.

Exercise 7: Exceptions (Chapter 8)

Write `safe_div(a, b)` that returns the quotient, or a message if `b` is 0 or the inputs are not numbers.

Exercise 8: Debug the keyword counter

The keyword-frequency script from the IT and Digital Culture file undercounts words that are followed by punctuation (for example, "culture."). Find the cause and fix the line that splits the post into words.

Part 2: More Python Exercises (Classes, Modules, File I/O)

Exercises

Classes 1: Shared versus instance variables (Section 9.3). Predict what `print(b.tricks)` shows, then fix the class so each dog has its own tricks.

```
class Dog:
    tricks = []
    def __init__(self, name):
        self.name = name
    def add_trick(self, t):
        self.tricks.append(t)

a, b = Dog("A"), Dog("B")
a.add_trick("roll")
print(b.tricks)
```

Classes 2: Inheritance (Section 9.5). Write an `Account` class with `owner`, `balance`, a `deposit(amount)` method that rejects amounts of 0 or less, and a `__str__` method. Then write `SavingsAccount` that inherits from it, adds an interest rate (default 0.02) and an `add_interest()` method. Create `SavingsAccount("Amina", 1000)`, deposit 500, add interest and print it.

Classes 3: Iterators and generators (Sections 9.8 to 9.10). Write a class `Reverse` that loops over a sequence backwards, then write the same thing as a generator function. Finally, compute the sum of the squares of 0 to 3 with a generator expression.

Modules 1: Creating a module (Section 6.1). Create a file `fibo.py` with a `fib(n)` function. Show three ways to import it from another file, and print `fibo.__name__`.

Modules 2: Running a module as a script (Section 6.1.1). Change `fibo.py` so that `python fibo.py 50` prints the Fibonacci numbers below 50, but importing it prints nothing.

File I/O 1: Reading and writing (Section 7.2). Write three lines to `notes.txt`, then read it back line by line without adding blank lines. Explain why `with` is recommended.

File I/O 2: JSON (Section 7.2.2). Save the dictionary `{"name": "Amina", "scores": [12, 15]}` to `d.json` and load it back. Handle the case where a file does not exist.

Part 3: Git and GitHub Exercises

Exercises

Git 1: Order the commands. You edited `file1` in your local repository. Put these commands in the correct order and say when you would add `git status`: `git push origin main`, `git commit -m "Fix typo"`, `git add file1`, `git pull`.

Git 2: Resolve a conflict. You and your partner both edited `file1`. Your partner pushed first. Your `git push origin main` is rejected. List the commands you run, and explain what happens to `file1` after `git pull`.

Git 3: Branching. Write the commands for this story: you start an experiment in a branch named `exp`, you are interrupted by an urgent bug in `file1` on `main`, you fix and push it, you bring the fix into `exp`, you finish the experiment, and you finally merge it into `main` and delete the branch.

Git 4: Mirror an assignment repository. Your course publishes starter code at `https://github.com/COS217/Survey`. Write the commands to create your own private copy named `My_COS217_A0` under your account `yourname`, then a working copy on your computer.

Git 5: Setup. Write the commands to set your name, email and editor, and to stop being asked for your credentials on every push. Why can you not use your GitHub password on the command line?

Git 6: Short questions. (a) What is the difference between the local repository and the GitHub repository? (b) Why are `git add` and `git commit` separate steps? (c) What does `git rm file1` do that deleting the file with your file manager does not? (d) Which two habits does the primer recommend to avoid conflicts?

Part 4: Machine Learning Exercises

Exercises

ML 1: Name the learning type. For each case, say whether it is supervised, unsupervised, semi-supervised, self-supervised or reinforcement learning: (a) a spam filter trained on emails flagged by users; (b) grouping blog visitors into similar groups; (c) a robot learning to walk by receiving rewards; (d) predicting a car's price from mileage and age; (e) a photo service that finds the same person across photos, then names them after you give one label; (f) masking part of an image and training a model to restore it.

ML 2: Compute. (a) A linear model has $\theta_0 = 3.75$ and $\theta_1 = 6.78 \times 10^{-5}$. Predict life satisfaction for a country with GDP per capita of $37,655. (b) With k-nearest neighbors (k = 3), the three closest countries have life satisfaction 7.2, 5.9 and 5.9. What is the prediction? (c) A spam filter classifies 170 of 200 test emails correctly. What is its accuracy?

ML 3: Diagnose the model. (a) Training error is 2% and test error is 18%. (b) Training error is 20% and test error is 21%. For each case, name the problem and give two fixes.

ML 4: Batch or online? Choose the better approach and justify it: (a) predicting stock prices that change every minute; (b) a rover on Mars with limited resources; (c) a dataset too large to fit in memory; (d) a cat versus dog classifier that changes slowly.

ML 5: Test set misuse. You trained 100 models with different hyperparameters and picked the one with 5% error on the test set. In production it makes 15% errors. What went wrong, and what should you have done?

ML 6: Code. The code below trains a linear model. Change it to use k-nearest neighbors with k = 3.

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X, y)
```

ML 7: Sampling bias. In 1936 the Literary Digest polled about 10 million people and predicted that Landon would win, but Roosevelt won. Name the two flaws in its sampling.